

PES2UG23CS678_VarunNamboodiri_T

DL_Assignment_Unit1

Name: Varun Namboodiri

Section: K

SRN: PES2UG23CS678

Date: 08-02-2026

Overview:-

1. Table 0.
2. Table 1.
3. Questions.

Table 0:-

This table is for the MNIST dataset.

CNN Architectures\ Parameters	No. of parameters	Training time	Final training & validation accuracy
LeNet5	62000	10-15 s	98-99%
AlexNet	23 million	30-45 s	99%
ResNet18	11 million	1 minute	99.5%

Table 1:-

This table is for the CIFAR-10 architecture

CNN Architectures \ Parameters	No. of parameters	Training time	Final training & validation accuracy
LeNet5	62000	15 s	55%
AlexNet	23 million	45 s	75%
ResNet18	11 million	1 min 5 sec	85%

Questions:-

Q1) Based on the different architectures given, calculate the total no. of parameters in each type of CNN.

Ans:- For LeNet5: 60-62 thousand parameters.

For AlexNet: 23 - 28 million parameters.

For ResNet18: 11 million parameters.

Q2) How does the number of parameters in each model correlate with accuracy on MNIST and CIFAR-10?

Ans:- On CIFAR-10: The correlation is strong. CIFAR-10 has complex features (color, varied backgrounds). LeNet lacks the capacity (too few parameters) to learn these features, resulting in low accuracy (~55%). ResNet and AlexNet perform significantly better due to higher capacity. On MNIST: The correlation is weak. Because MNIST is a simple dataset, LeNet (60k params) performs almost as well as ResNet (11M params). Extra parameters in ResNet are "overkill."

Q3) How does dataset complexity (grayscale vs. RGB) impact model accuracy and training time?

Ans:- Grayscale reduces the training time, for the training of the grayscale images are comparatively easier in nature. Here, it would be hard to do a refined feature detection in complicated images.

RGB increases the training time, for there are more colours to map, thereby making it harder to train these images. But, it would be easy to do a refined feature detection in complicated images due to more colours being present as opposed to the grayscale images.

Q4) What trade-offs exist between accuracy and computational efficiency?

Ans:- Modern architectures such as ResNet demonstrate that using the help of architectural innovations (like residual learning and the removal of heavy fully connected layers), it is possible to achieve higher accuracy with fewer parameters and refined computational efficiency than the older, bulkier predecessors such as AlexNet.

Henceforth, the goal of CNN evolution is to move towards achieving maximum accuracy twinned with minimum computational cost.